ARCHIE BENN

HOW TO TRAIN YOUR MODEL

ASSESSING THE IMPACTS OF FOREST AGE ON THE SPATIAL AND TEMPORAL GENERALISATION OF RANDOM FOREST EVAPOTRANSPIRATION MODELS

MSc Bioinformatics
Supervisor: Martin De Kauwe
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ABSTRACT

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Key words: Evapotranspiration, Forest Age, Machine Learning, FLUXNET. **Word count:** 4000

DECLARATION

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn

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A statement requiring citation [1].

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2.1 Data Acquisition and Setup

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3.1 Paragraphs

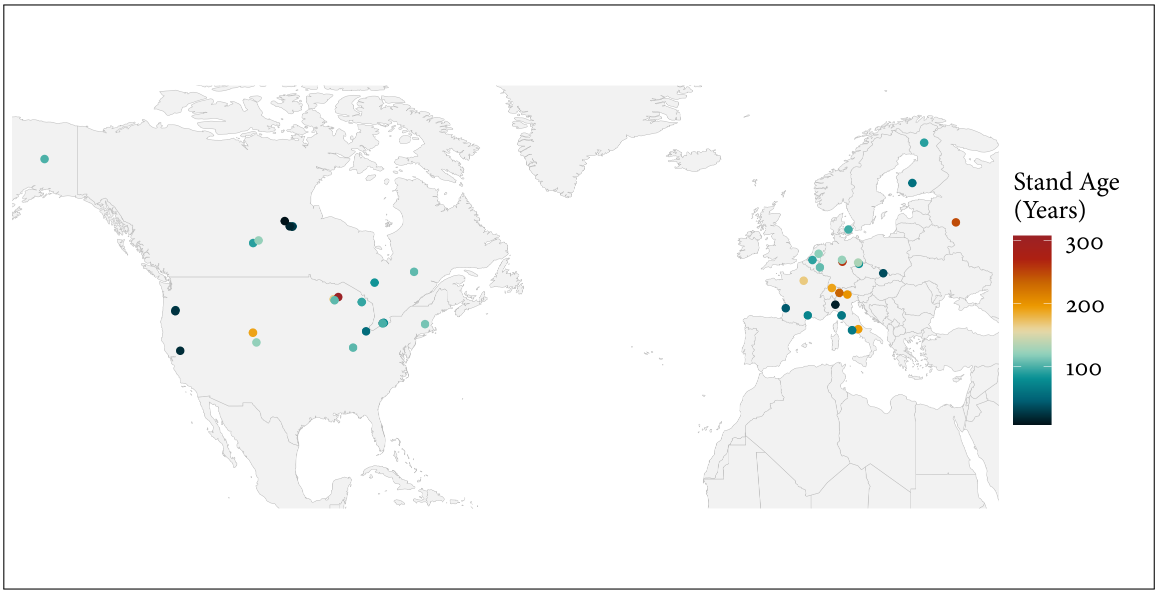

Figure 1: Map of FLUXNET sites and associated forest stand age in this study

Table 1: Main characteristics of each cluster defined during cluster analysis. To see all clusters on a map see Fig. X (supplementary materials).

Cluster	Medoid Site	Defining Characteristics
1	DE-Lnf	Temperate sites without significant features.
2	US-UMd	Continental sites without significant features.
3	CA-Obs	Continental sites with low mean air temperatures.
4	CA-TP3	Continental sites with high precipitation.
5	IT-SRo	Temperate sites with high air temperatures.
6	US-Me3	Temperate sites with very high VPD, high SW radiation, low stand age, low wind speeds, and low precipitation.

3.2 Cluster Analysis

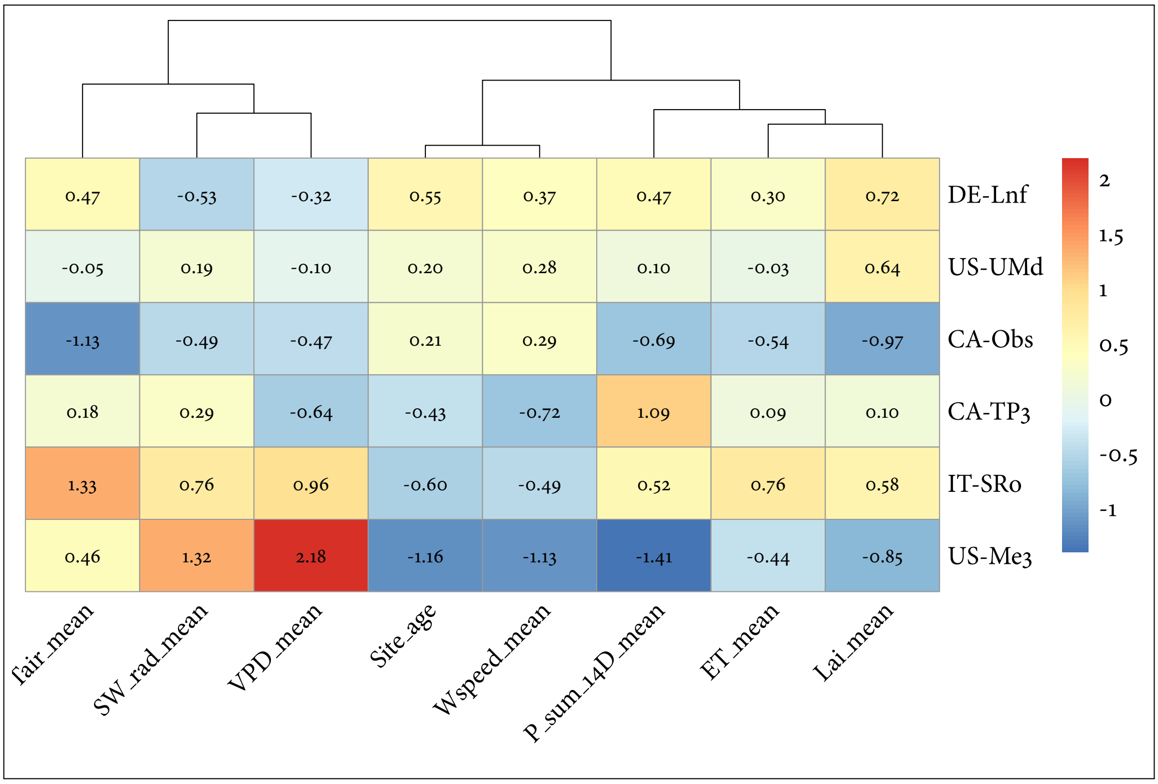

Figure 2: Pheat

3.3 Spatial Generalisation

Suggested framing:

DE-Tha (age helps) vs CA-Obs (age hurts) as your case studies For each: SHAP dependence plot for age, feature importance ranking, brief site characteristics (climate, age, cluster membership) Then the permutation test as a cross-cutting check — not per-site, but run once (or per-cluster) across the whole train/test setup to test whether age's apparent value is real signal or site-proxy leakage

Mention the other 4 clusters exist and were tested (maybe one summary table/plot of all 6 site R^2 deltas) so it's clear you didn't cherry-pick — then justify the deep dive as case studies. That's a normal, well-scoped move for a 4k-word section, not weak coverage.

Yeah, that's a sensible scope for 4k words total. Rough shape:

Spatial (bulk of results): cluster analysis overview (brief, all 6), then deep dive on DE-Tha vs CA-Obs (SHAP, feature importance, why age helps/hurts), permutation test as the credibility check Temporal: shorter section — does age still help/hurt over time at a couple of long-record sites, tying back to the leakage question (age varies within-site here, so it's a cleaner test) Discussion: synthesize — when/why age helps, leakage risk, what it means for future FLUXNET+age model building

3.3.1 *Analysis 1.1*

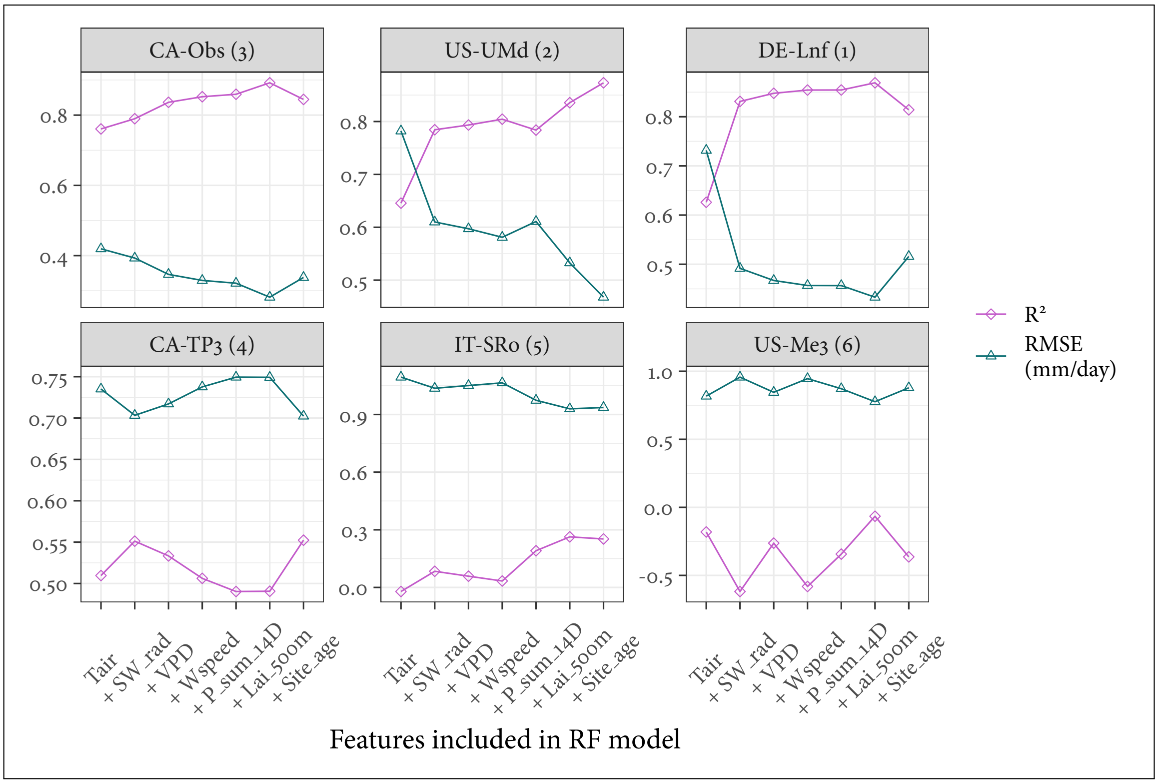

Figure 3: R^2 and RMSE for medoid cluster site predictions of ET, ordered by peak R^2 with clusters indicated in facet names. Models built and tested using a consecutively greater number of input features from FLUXNET and LAI data. For example, the '+VPD' model incorporates Tair, SW_rad, and VPD as input features.

3.3.2 *Analysis 1.2*

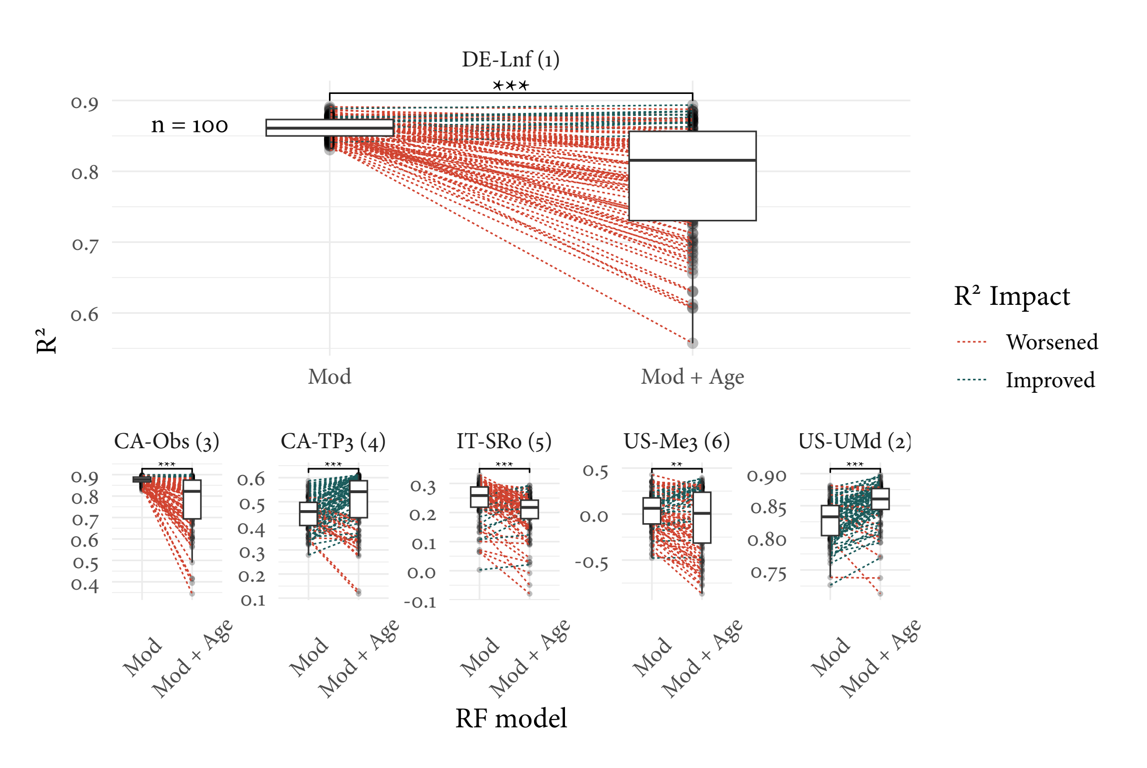

Figure 4: Paired box-plots indicate how R^2 changes when age is added as a feature to RF models across 100 repeated runs of randomly sampled training datasets. Points are connected on a per-run basis where training dataset site composition is identical. Wilcoxon signed-rank tests suggest model performances differ significantly at all test sites. DE-Lnf is shown large for readability, with other cluster medoid sites shown at a reduced scale for comparison.

3.4 Temporal Generalisation

4 DISCUSSION

5 CONCLUSION

REFERENCES

[1] A. J. Figueredo and P. S. A. Wolf. Assortative pairing and life history strategy - a cross-cultural study. *Human Nature*, 20:317–330, 2009.

6 SUPPLEMENTARY MATERIAL